MSCF Probability Exercises

Solutions to all of these exercises are provided in a separate document.

1. Show that if two events A and B are independent, then A^c and B^c are also independent.

2. Assume that $P(B) > 0$. Show that $P(A \mid B) = 1 - P(A^c \mid B)$.

3. A health maintenance organization (HMO) concerned with the adverse financial effects of catastrophic medical expenses for its members enters into a reinsurance arrangement in which the reinsurer (i.e., the insurer providing insurance to the HMO) agrees to pay the amount of each plan member's annual hospital expenses that exceed \$100,000. Thus, for example, if the HMO had exactly two members and these two members had annual expenses of \$130,000 and \$70,000, respectively, then the reinsurer would pay the HMO \$30,000. Under the terms of the agreement, the HMO will be required to pay additional reinsurance premium if the fraction of the plan membership for which the reinsurance claim is filled exceeds 5%. This condition helps minimize any incentive the HMO may have to increase its enrollment of substandard risks after entering into the reinsurance arrangement.

   The probability of catastrophic claim for any given member clearly depends on many factors, one of which is the age of the member. On the basis of historical data, the plan estimates that 10% of members over the age of 65 will incur hospital expenses exceeding \$100,000, 4% of members between the ages of 40 and 65 will incur hospital expenses exceeding \$100,000, and 3% of members under the age of 40 will incur hospital expenses exceeding \$100,000.

   Twenty percent of the HMO's members are currently over the age of 65. If the HMO wishes to maintain this level of members over the age of 65 (perhaps for competitive or regulatory reasons) and wishes to avoid exceeding the 5% trigger point in the reinsurance agreement, how large must the HMO's under-40 population be?

   Hint: Let C be the event that a member has hospital expenses exceeding \$100,000, use a Law of Total Probability and solve $P(C) \leq .05$.

4. A box contains three coins. Two of these are fairly unusual coins: one has heads on both sides, one has tails on both sides. The other is a fair coin.

   One of the three coins is chosen randomly (with each coin equally likely) and then it is tossed once. The toss comes up heads. What is the probability that the coin chosen is the fair coin?

5. Suppose that events A_1 and A_2 are independent conditional on event B.
   Show that $P(A_2 \mid A_1 \cap B) = P(A_2 \mid B)$. You can assume that $P(A_1 \cap B) > 0$.

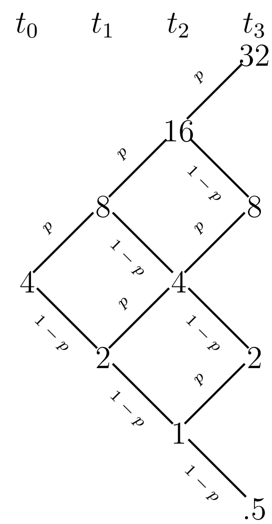
6. Let X be a discrete random variable whose possible values are the nonnegative integers. The probability mass function (pmf) for X satisfies the relationship

$$f_X(k) = \frac{1}{2} f_X(k-1), \quad k = 1, 2, \ldots$$

What is the pmf for X?

7. The *binomial model* for stock prices describes the evolution of prices in discrete time. If the price at time t is S_t, the value at time $t+1$ will be uS_t with probability p or dS_t with probability $1-p$. Under this model the steps are independent from one time period to the next.

When $d = 1/u$ the model takes a particularly simple form and can be depicted using a tree. For example, if $S_0 = 4$, $u = 2$, and $d = 1/2$, then the model has the following structure:



In the case where $S_0 = 4$, $u = 2$, and $d = 1/2$, and using $p = 0.4$, find $P(S_7 \geq 32)$.

8. Let $X \sim$ Geometric(p) with $p < 1$. Prove that X has the *memoryless property*: For $n = 0, 1, \ldots$ and $m = 0, 1, 2, \ldots$, it holds that

$$P(X \geq n + m \mid X \geq n) = P(X \geq m)$$

Comment: This is called the "memoryless property" because the above equality can be interpreted as "If one knows that the first n trials were failures (i.e., that $X \geq n$), then the probability that there are at least m additional failures (that $X \geq n + m$) is identical to the probability there were at least m failures from the outset (that $X \geq m$)." The geometric distribution is the only discrete distribution with the memoryless property.

9. Suppose that a 10 ml sample of swimming pool water is declared contaminated if it is found to have at least two bacteria. It is reasonable to assume that the number of bacteria in such a sample is well-modelled using the Poisson distribution with mean $\lambda = 1.4$.

   If 20 independent 10 ml samples of pool water are obtained, what is the probability that exactly half of them are contaminated?

10. This exercise proves an often seen "trick" for calculating expected values in a particular situation, and then illustrates its use.

    (a) For a nonnegative integer-valued random variable X, show that

    $$\sum_{i=1}^{\infty} P(X \geq i) = E(X).$$

    (b) Let $X \sim \text{Geometric}(p)$. Use part (a) to prove that $E(X) = \dfrac{1-p}{p}$.

11. Let the CDF of a continuous random variable X be as follows:

    $$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - \displaystyle\sum_{k=0}^{2} \frac{(2x)^k \, e^{-2x}}{k\,!}, & x \geq 0 \end{cases}$$

    Determine the pdf of X, $f_X(x)$. What "named" distribution is this?

12. Let X have the pdf

    $$f_X(x) = 0.5e^{-|x|} \qquad -\infty < x < \infty.$$

    Find $P(2.5 \leq |X| \leq 3.5)$.

13. The length of the life of an electronic component is adequately modelled using the exponential distribution with mean of 10000 hours. Given that the component has been working for 9000 hours, what is the probability that it will work for no more than 11000 hours? Repeat under the assumption that the lifetime (in hours) is not exponential but rather uniformly distributed over (8000, 12000).

14. This is a continuous version of Exercise 10 above.

    (a) For a nonnegative continuous random variable Y, show that

    $$\int_0^{\infty} P(Y > t)\, dt = E(Y).$$

    (b) Let X have the exponential(λ) distribution. Use part (a) to show that $E(X) = 1/\lambda$.

15. Let X be a continuous random variable, let m be a constant, and define $g(X) = \min(X, m)$.

    (a) Show that
    $$E[g(X)] = \int_{-\infty}^{m} x f_X(x)\, dx + m\,[1 - F_X(m)]\,.$$

    (b) Now assume X is nonnegative and that let m is also nonnegative. Show, using integration by parts and part (a), that
    $$\int_{0}^{m} [1 - F_X(x)]\, dx = E[g(X)]\,.$$

    Comment: Note the similarity between this result and the expressions derived in Exercises 10 and 14.

    (c) Consider an insurance policy with a **cap m** where the insurer will compensate the policyholder for losses up to a maximum of m. In this case, if the random variable X represents the loss (or claim) size then the payment of the insurer is $\min(X, m)$.

    An insurance company offers insurance against a particular type of loss. The particular policy has a cap but no deductible. Losses are assumed to have an exponential distribution with mean of $2000. At what level should the cap be set if the insurer would like the expected payment for any loss of this type to be $1500?

16. Suppose that $X \sim \text{exponential}(\lambda_1)$ and $Y \sim \text{exponential}(\lambda_2)$, and that X and Y are independent. What is the distribution of $\min(X, Y)$?

17. Let $f(x)$ and $F(x)$ be the density and CDF, respectively, of a continuous distribution. Prove that the function $g(x)$, defined below, is also a legitimate density.
    $$g(x) = 2F(x)f(x), \quad -\infty < x < \infty$$

    Hint: Consider the substitution $u = F(x)$.

18. **The St. Petersburg Paradox.** A person tosses a fair coin until tails appears for the first time. If tails appears on the n^{th} flip, the person wins 2^n dollars. Let X denote the player's winnings.

    (a) Show that $E(X) = \infty$.

    (b) Suppose the price of playing this game was $1 million. What is your probability of coming out ahead? Would you be willing to pay $1 million to play this game?

19. There are two **known** possible causes for the breakdown of a machine. The cost of checking for the first cause is C_1 dollars, and if that were the cause for a breakdown, the cost of the repair would be R_1 dollars. Similarly, there are costs C_2 and R_2 associated with the second cause. In the past, the proportion of breakdowns that were from the first

cause was p, and the proportion of breakdowns from the second cause was $1-p$. (Since it is always possible that there are additional causes, a diagnostic check must be performed prior to making a repair, even if the other cause has been ruled out.)

Under what conditions on p, C_i, R_i, for $i = 1, 2$, should we check the first possible cause of breakdown and then the second, as opposed to reversing the checking order, so as to minimize the expected cost involved in returning the machine to working order?

Note: You can assume that it is not possible that both causes of breakdown have occurred.

20. A man claims to have extrasensory perception (ESP). As a test, a fair coin is flipped ten times and the man is asked to predict the outcome in advance, before each flip. He gets 8 out of 10 correct. What is the probability that he would have done *at least* as well just by chance (i.e., if he had no ESP)?

21. A family of distributions that has been used to model income, city population size, and size of firms, among other things, is the **Pareto**. There are two positive parameters, k and θ, and the pdf is given as follows:

$$f_X(x) = \begin{cases} \dfrac{k\theta^k}{x^{k+1}}, & x \geq \theta \\ 0, & \text{otherwise} \end{cases}$$

   (a) Verify that this is a legitimate family of pdfs.
   (b) What is the CDF for this distribution?
   (c) Find an expression for the median of the Pareto family.

22. Suppose that X is a positive random variable. We know that $E(1/X) \neq 1/E(X)$, in general. Can you claim that either $E(1/X) \geq 1/E(X)$ or $E(1/X) \leq 1/E(X)$? Explain your response.

23. Consider a circle of radius 1 centered at the origin, and suppose that a point within the circle is randomly chosen in such a manner that all regions within the circle of equal area are equally likely to contain the point. (In other words, the point is uniformly distributed within the circle.) If we let X and Y be the coordinates of the point chosen, it follows that the joint density function of X and Y is given by:

$$f_{XY}(x, y) = \begin{cases} 1/\pi & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find the marginals of X and Y. (They are the same due to symmetry.)

(b) Let D be the distance between the origin to the point selected. Find the probability that D is less or equal to d. In other words, find $F_D(d)$.
Guidance: Note that $D = \sqrt{X^2 + Y^2}$.

(c) Find $E(D)$.

24. Consider the following joint pdf:

$$f_{X,Y}(x,y) = \begin{cases} ye^{-y(x+1)} & x > 0, \quad y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the density function of $T = XY$, and identify the distribution you got.

Guidance: Find $F_T(t) = P(XY \leq t) = P(Y \leq t/X) = \int\int \ldots$ and then differentiate.

Note: One of the two ways to setup the double integration is more "friendly" than the other.

25. Assume that U_1 and U_2 are independent Uniform$(0, 1)$ random variables. Let $U_{(1)}$ denote the minimum of U_1 and U_2, and let $U_{(2)}$ denote the larger of those two. Finally, let

$$L_1 = U_{(1)}, \quad L_2 = U_{(2)} - U_{(1)}, \quad \text{and} \quad L_3 = 1 - U_{(2)}.$$

Comment: This is the "stick-broken-into-three-pieces" example, with L_i denoting the lengths of the three pieces ordered from left to right.

In lecture we derived the joint distribution of $(U_{(1)}, U_{(2)})$. You can build on that result here.

(a) What is the joint distribution (pdf) of (L_1, L_2)?
Guidance: Use the joint pdf of $(U_{(1)}, U_{(2)})$ as a starting point. Look at the support for $(U_{(1)}, U_{(2)})$ and ask: For what region is $L_1 \leq \ell_1$ and $L_2 \leq \ell_2$?

(b) Show that L_1 and L_2 both have the Beta$(1, 2)$ distribution. (Do not take it as "obvious" that these have the same distribution. Show the derivation.)

26. Let $X \sim \text{Exp}(\lambda_1)$ and that $Y \sim \text{Exp}(\lambda_2)$.

If we assume that X and Y are independent, then the joint pdf of X and Y is

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) = \begin{cases} \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y}, & x > 0,\ y > 0 \\ 0, & \text{otherwise} \end{cases}$$

(a) For $t > 0$, prove that $P(Y < tX) = \dfrac{t\,\lambda_2}{\lambda_1 + t\,\lambda_2}$

(b) For simplicity, assume that $\lambda_1 = \lambda_2 = 1$, and let $T = \dfrac{Y}{X}$.
Use part (a) to find the pdf of T, $f_T(t)$.

(c) Without doing any formal derivations beyond what was done in part (b), explain why $E(T) = \infty$.

27. Let (X, Y) have the following joint pdf:

$$f_{XY}(x, y) = \begin{cases} 2x, & 0 < x < 1, \ x < y < x + 1 \\ 0, & \text{otherwise} \end{cases}$$

Find $f_X(x)$ and $f_Y(y)$.

Comment: People often ask: "Is there a mistake here? The pdf does not depend on y." There is no mistake; think about how the pdf does depend on y.

28. An auto insurer is developing a model for claim size in an attempt to better price its products. On the basis of historical data, it has determined that the claim size distribution for policy holders classified as good risks has an exponential distribution with *mean* 1/2 and the claim size distribution for policy holders classified as bad risks has an exponential distribution with *mean* 3, where claim sizes are measured in thousands of dollars. Ignoring all other factors, there is 30% chance that a policyholder is a bad risk.

Let X be the claim size, and let Y be the Bernoulli($p = 0.30$) random variable which takes the value 1 if the policyholder is bad risk and 0 otherwise. In other words,:

$$Y = \begin{cases} 1 & \text{with probability } 0.30 \\ 0 & \text{with probability } 0.70 \end{cases}$$

(a) Use all of this information to show that the marginal distribution of X is:

$$f_X(x) = 1.4e^{-2x} + 0.1e^{-x/3} \quad x > 0.$$

Comment: In this example, the claim size distribution function is a weighted sum of some other distribution functions. To be precise:

$$F_X(x) = (0.70)F_{X_1}(x) + (0.30)F_{X_2}(x)$$

where X_1 is the random variable with density $f_{X_1}(x) = 2e^{-2x}$ and X_2 is the random variable with density $f_{X_2}(x) = \frac{1}{3}e^{-x/3}$. A random variable with this property is referred to as a *discrete mixture.*

In general, a discrete mixture is a random variable X whose distribution function has the form

$$F_X(x) = p_1 F_{X_1}(x) + p_2 F_{X_2}(x) + \cdots + p_n F_{X_n}(x).$$

where $0 < p_i < 1$ for all i and $\sum p_i = 1$. Mixtures arise in connection with the Law of Total Probability and are extremely important in insurance applications where the risk class of a subject is uncertain.

(b) What is the probability that a policyholder's claim exceeds $1,000?

29. Let X and Y be independent random variables each taking the values +1 and -1 with probability 0.5. Define

$$U = XY \quad \text{and} \quad V = X + Y.$$

   (a) Write down a table giving the joint probability mass function of U and V.

   (b) Use the values in your table to determine $E(U)$, $E(V)$, and the covariance between U and V.

   (c) Are U and V independent?

30. Let (X, Y) have the following joint pdf:

$$f_{XY}(x,y) = \begin{cases} 60x^2y & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find the marginal distributions $f_X(x)$ and $f_Y(y)$.

   (b) Use properties of the Beta distribution to find $E(X)$, $E(Y)$, $V(X)$, and $V(Y)$.

   (c) Find $\text{Cov}(X, Y)$ and $\text{Corr}(X, Y)$.

   (d) Find $V(X + Y)$.

31. The proportion of defective products coming out of a production line, P, is a random variable, and varies day to day according to a Beta distribution with parameters $\alpha = 1$ and $\beta = 9$. In other words, each day has its own value of P, drawn from the specified distribution. On a randomly chosen day 100 products are sampled and inspected. Let X be the number of defective products out of 100. Find $E(X)$ and $V(X)$.

32. Suppose that X_1, X_2, X_3 have the following joint pdf:

$$f_{X_1X_2X_3}(x_1, x_2, x_3) = \begin{cases} 6\, e^{-(x_1+2x_2+3x_3)} & x_i > 0 \ \ ,\ i = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

   Are X_1, X_2, and X_3 independent? Justify your answer rigorously.

33. Consider the following situation:

   - $Y \sim N(\mu, \sigma^2)$
   - $(X \mid Y = y) \sim N(y, \tau^2)$ [FYI: τ is the Greek letter Tau]

   We want to find the distribution of X.

One strategy would be to find $f_{XY}(x, y)$ and then use a Law of Total Probability

$$f_X(x) = \int_y \underbrace{f_{X|Y}(x \mid y) f_Y(y)}_{f_{XY}(x,y)} \, dy$$

While this is fine, it requires some algebraic manipulations that may not be obvious. Another strategy is to find the MGF $M_X(t)$ and use it to identify the distribution of X:

$$M_X(t) = E(e^{tX}) = E[E(e^{tX}|Y)]$$

where the "inner" expected value, $E(e^{tX}|Y)$ is the moment generating function of the random variable $X|Y = y$.

Find $M_X(t)$ and use to determine the distribution of X.

34. Suppose that X_1, X_2, X_3 have the following joint pdf:

$$f_{X_1 X_2 X_3}(x_1, x_2, x_3) = \begin{cases} \frac{1}{3}(x_1 + 2x_2 + 3x_3) & 0 < x_i < 1,\ i = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the joint pdf of (X_1, X_2).

(b) Find $P\left(X_3 < \frac{1}{2} \,\middle|\, X_1 = \frac{1}{4},\ X_2 = \frac{3}{4}\right)$.

35. Assume that X, Y, and Z are all random variables. Use our measure-theoretic notion of conditional expectation to explain why

$$E(Y|X) = E(E(Y|X, Z)|X).$$

36. Assume that $\mathcal{F}$ is a σ-field and X is a random variable. Show that

$$E(XE(X|\mathcal{F})) = E(E(X|\mathcal{F})^2).$$

37. If X has the $N(\mu, \sigma^2)$ distribution, then $Y = e^X$ has the lognormal(μ, σ^2) distribution. Find an expression for $E(Y^k)$, the k^{th} moment of Y, for $k = 1, 2, \ldots$.

38. The random variables X and Y have a joint density function given by

$$f_{XY}(x, y) = \begin{cases} 2\,e^{-2y}/y & 0 < y,\ 0 < x < y \\ 0 & \text{otherwise} \end{cases}$$

Compute $\text{Cov}(X\,, Y)$

Hint: To find $E(X)$, note that finding $f_X(x)$ is not so easy. Instead, use the Rule for the Lazy Statistician directly on $f_{XY}(x, y)$, i.e., with $g(x, y) = x$.

39. Suppose that $X_1, X_2, \ldots, X_n$ are iid, each with mean μ and variance σ^2. Assume that $\mu \neq 0$ and $0 < \sigma^2 < \infty$. Approximate the distribution of $(\overline{X})^3$. You can assume that n is larger than 100.

40. An insurance company has 10,000 automobile policyholders. The expected yearly claim per policyholder is \$240 with a standard deviation of \$800. Approximate the probability that the total of all yearly claims exceeds \$2.5 million.

41. A certain component is critical to the operation of an electrical system and must be replaced immediately upon failure. If the mean lifetime of this type of component is 100 hours and its standard deviation is 30 hours, how many of these components must be in stock so that the probability that the system is in continual operation for the next 4000 hours is at least 0.95?

42. Let $\mathbf{X}$ be multivariate normal: $\mathbf{X} \sim \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 1 \end{bmatrix}.$$

(a) What is the joint distribution of (X_2, X_3)? Are X_2 and X_3 independent?

(b) Find $E(X_2^2 \cdot e^{X_3})$

(c) Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

What is the distribution of $\mathbf{Y} = \mathbf{AX}$? Are the components of $\mathbf{Y}$ independent of each other?